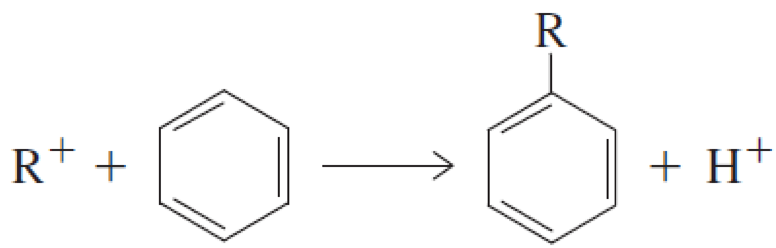
Qualitative Analysis Separation

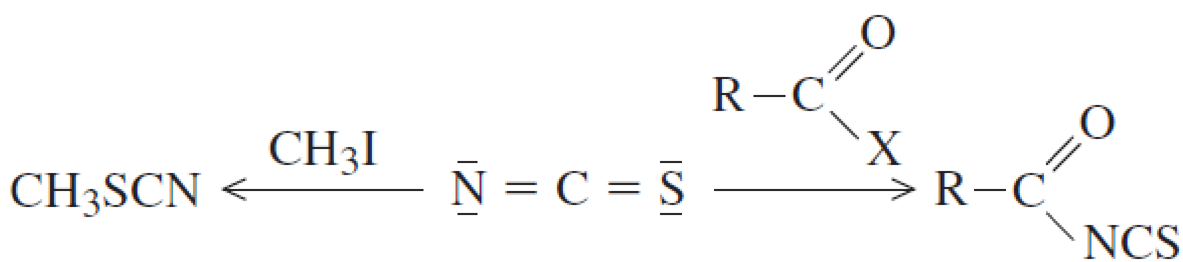
	Group 1	Group 2	Group 3	Group 4	Group 5
HSAB acids	Soft	Borderline and soft	Borderline	Hard	Hard
Reagent	HCl	H_2S (acidic)	H_2S (basic)	$(NH_4)_2CO_3$	Soluble
Precipitates	AgCl	HgS	MnS	$CaCO_3$	Na^+
	$PbCl_2$	CdS	FeS	$SrCO_3$	K^+
	Hg_2Cl_2	CuS	CoS	$BaCO_3$	$NH_4{}^+$
		SnS	NiS		
		As_2S_3	ZnS		
		Sb_2S_3	$Al(OH)_3$		
		Bi_2S_3	$Cr(OH)_3$		

The function of the acid catalyst is to generate a *positive* attacking species. In the *Friedel-Crafts reaction*, $AlCl_3$ functions as an acid catalyst by increasing the concentration of the carbocation, R^+, which is the attacking species as shown in the following equations:

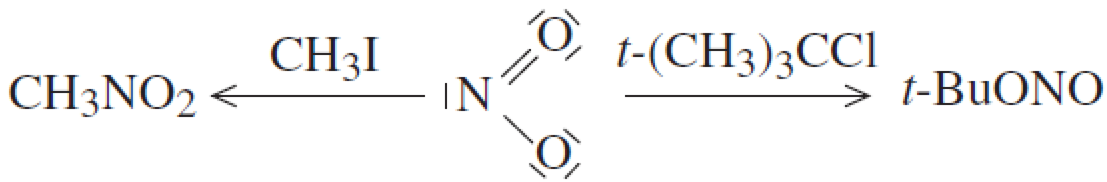
$$RCl + AlCl_3 \leftrightarrows R^+ + AlCl_4^-$$



Reactive site preference. We have already used the HSIP principle as it applies to linkage isomers in metal complexes. This application to bonding site preference can also be used to predict the products of some reactions. For example, the reactions of organic compounds also obey the principle when reacting with nucleophiles such as SCN^- or NO_2^-. Consider the reactions illustrated in Eq. (5.41):



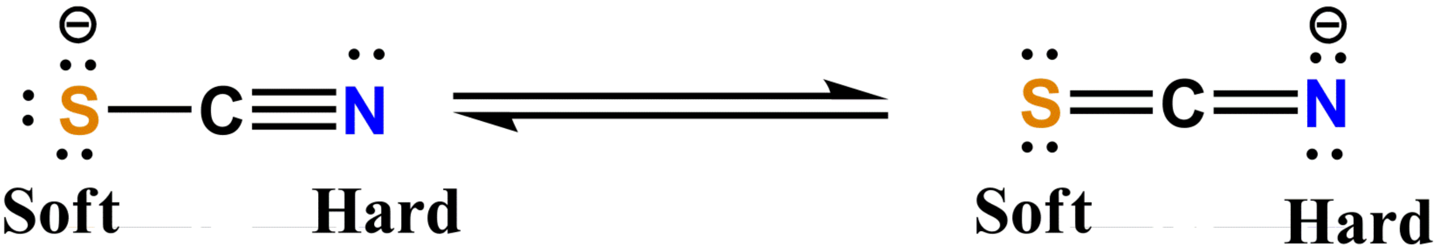
In this case, the acyl group in the acidic species RCOX is a hard acid and reacts with the nitrogen end of SCN^- to form an acyl isothiocyanate. The soft methyl group in methyl iodide bonds to the S atom and forms methyl thiocyanate. Consider the following reactions of NO_2^-:



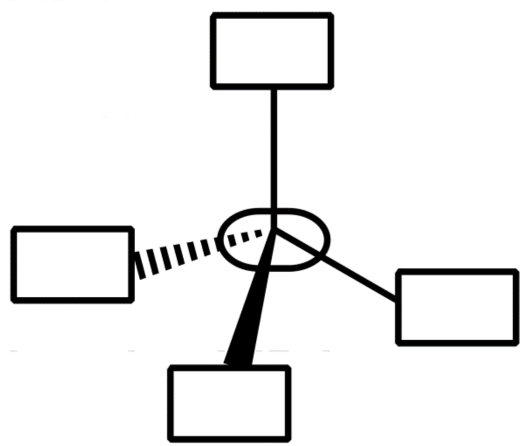
Here, the t-$(CH_3)_3C^+$ carbocation is a hard Lewis acid, so the product is determined by its interaction with the oxygen (harder) electron donor in NO_2^-. In the reaction with CH_3I, the product is nitromethane, showing the softer character of the methyl group.

Another reaction that illustrates the HSIP is the reaction of PCl_3 with AsF_3:

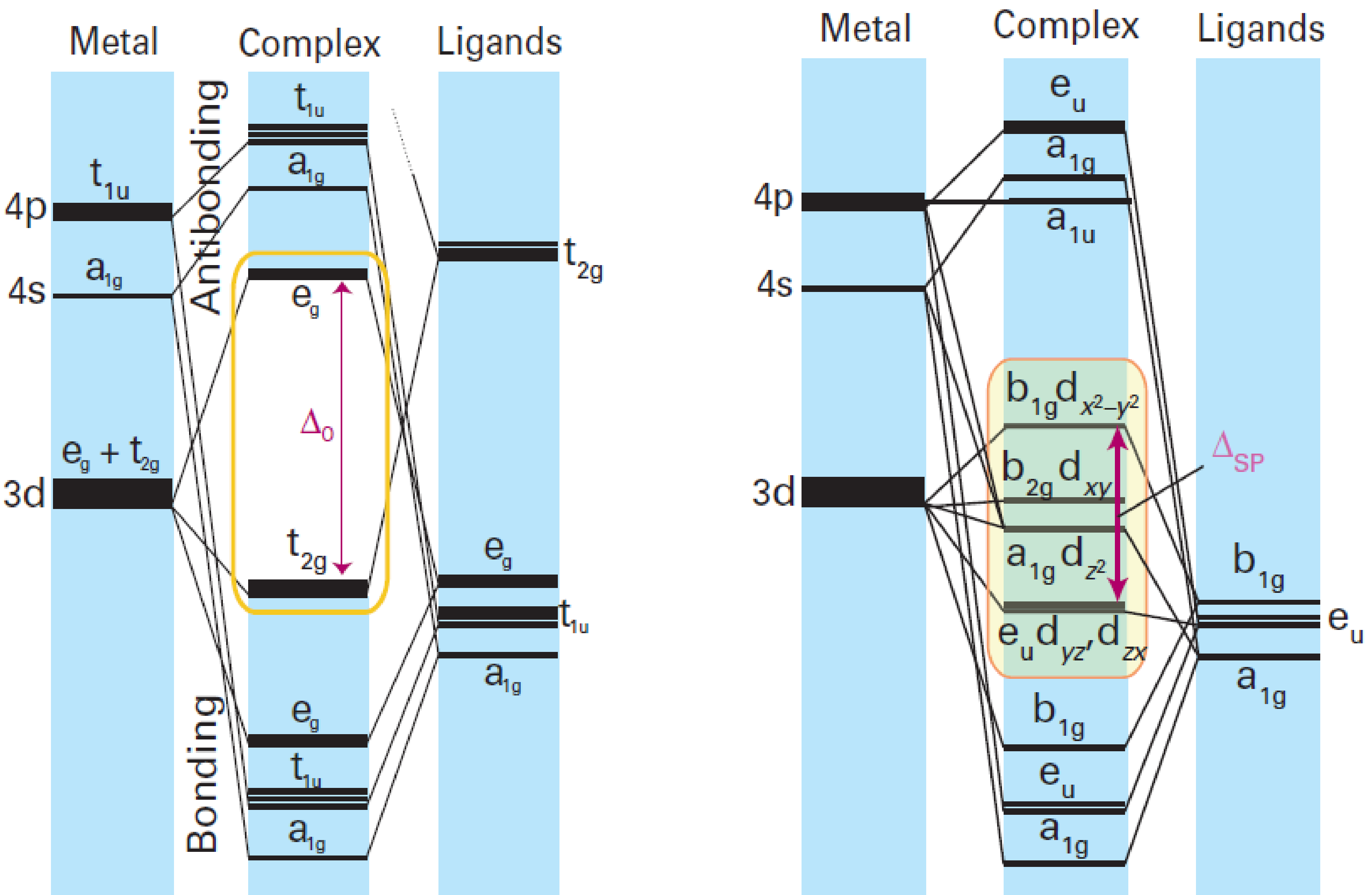
$$PCl_3 + AsF_3 \rightarrow PF_3 + AsCl_3$$



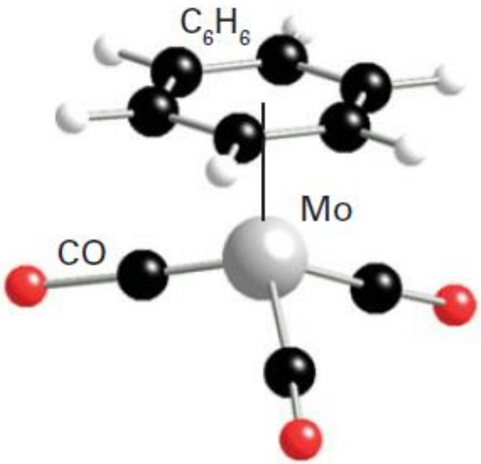
Zn^{2+} Hg^{2+}



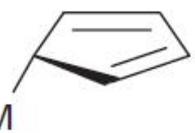
18-Electron Rule



Metal
Complex
Ligands
Antibonding
Bonding
t_{1u}
a_{1g}
4p
4s
3d
$e_g + t_{2g}$
e_g
t_{2g}
Δ_O
t_{1u}
a_{1g}
e_g
t_{1u}
a_{1g}
t_{2g}
e_g
t_{1u}
a_{1g}
Metal
Complex
Ligands
4p
4s
3d
e_u
a_{1g}
a_{1u}
$b_{1g}\, d_{x^2-y^2}$
$b_{2g}\, d_{xy}$
$a_{1g}\, d_{z^2}$
$e_u\, d_{yz}, d_{zx}$
Δ_{SP}
b_{1g}
e_u
a_{1g}
b_{1g}
e_u
a_{1g}



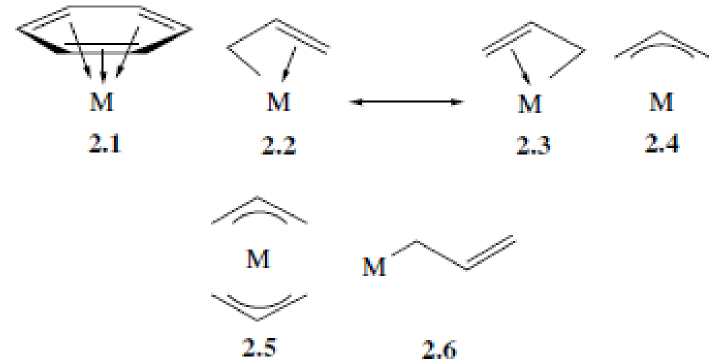
11 [Mo(η^6-C_6H_6)$(CO)_3$]

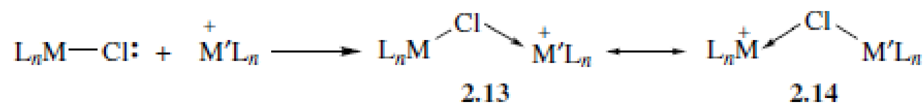


12 η^1-Cyclopentadienyl

13 η^3-Cyclopentadienyl

14 η^5-Cyclopentadienyl





(2.1)

THE ORGANOMETALLIC CHEMISTRY OF THE TRANSITION METALS

Fourth Edition

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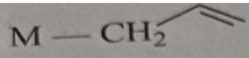
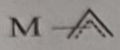
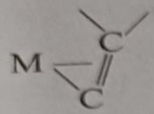
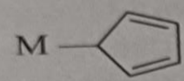
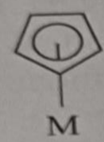
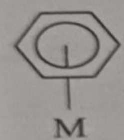
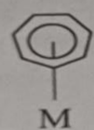
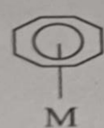
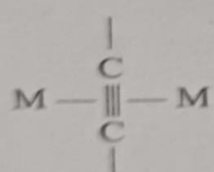
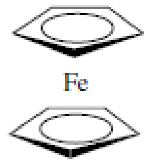
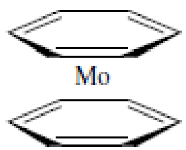
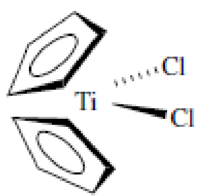
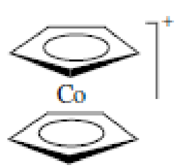
	Neutral atom electron count	Oxidation state electron count
Terminal ligands		
Carbonyl (M—CO)	2	2
Thiocarbonyl (M—CS)	2	2
Phosphine ($M—PR_3$)	2	2
Amine ($M—NR_3$)	2	2
Dinitrogen (M—N≡N)	2	2
Dihydrogen M—$\begin{matrix}H\\ \vert\\ H\end{matrix}$	2	2
Alkene M—$\begin{matrix}C\\ \Vert\\ C\end{matrix}$	2	2
Alkyne* M—$\begin{matrix}C\\ \vert\vert\vert\\ C\end{matrix}$	2	2
Isocyanide (M—CNR)	2	2
Nitrosyl, bent (M—$\ddot{N}$=O)	1	2
Nitrosyl, linear (M—N≡O)	3	2
Halogen (M—X)	1	2
Hydrogen (M—H)	1	2
Alkyl (M—R)	1	2
Acyl (M—C(=O)—R)	1	2
Aryl (M—Ph)	1	2
Amide ($M—NR_2$)	1	2
Phosphide ($M—PR_2$)	1	2
Alkoxide (M—OR)	1	2
Thiolate (M—SR)	1	2
Carbene = alkylidene ($M=CR_2$)	2	4
Carbyne = alkylidyne (M≡CR)	3	6

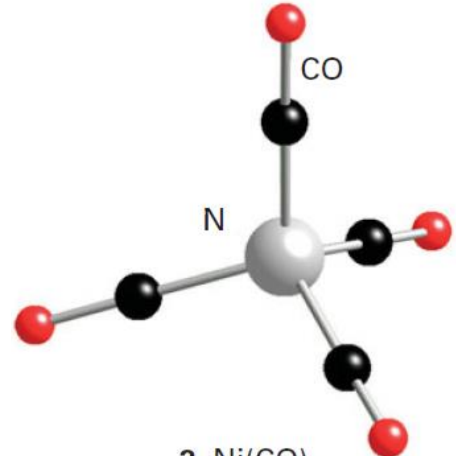
η^1-Allyl 	1	4
η^3-Allyl 	3	4
η^3-Enyl 	3	2
η^1-Cyclopentadienyl 	1	6
η^5-Cyclopentadienyl 	5	6
η^6-Benzene 	6	6
η^7-Cycloheptatrienyl 	7	6
η^8-Cyclooctatetraenyl 	8	10
Bridging ligands		
Carbonyl [M—(CO)—M]	2	2
Halogen (M—X—M)	3	4
Alkyne 	4	4
Hydrogen (M—H—M)	1	2
Alkyl [M—(CR_3)—M]	2	2
Amide [M—(NR_2)—M]	3	4
Phosphide [M—(PR_2)—M]	3	4
Alkoxide [M—(OR)—M]	3	4

0e	1e	2e	3e	4e
H^+	$H\bullet$[a]	$H^-(LiAlH_4)$[b]	NO	$C_3H_5^-(C_3H_5MgBr)$
$Me^+(MeI)$	$Me\bullet$[a]	$Me^-(LiMe)$		Butadiene
$Br^+(Br_2)$[c]		PPh_3, NO^+		NO^-
		Cl^-, CO, H_2		

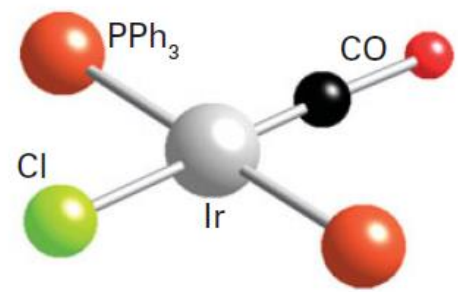
[a] These species are unstable and so they are invoked as reactive intermediates in mechanistic schemes, rather than used as reagents in the usual way.
[b] The reagents in parentheses are the ones most commonly used as a source of the species in question.
[c] Br_2 can also be a source of $Br\bullet$, a 1e reagent, as well as of Br^+, depending on conditions.

Ionic Model			Covalent Model	
$C_5H_5^-$ $C_5H_5^-$ Fe^{2+}	6e 6e 6e 18e	 **2.7**	$C_5H_5\bullet$ $C_5H_5\bullet$ Fe	5e 5e 8e 18e
Mo^{4+} $4 \times H^-$ $4 \times PR_3$	2e 8e 8e 18e	$MoH_4(PR_3)_4$ **2.8**	Mo $4 \times H\bullet$ $4 \times PR_3$	6e 4e 8e 18e
Ni^{2+} $2 \times C_3H_5^-$	8e 8e 16e	 **2.9**	Ni $2 \times C_3H_5\bullet$	10e 6e 16e
Mo $2 \times C_6H_6$	6e 12e 18e	 **2.10**	Mo $2 \times C_6H_6$	6e 12e 18e
$2 \times Cl^-$ Ti^{4+} $2 \times C_5H_5^-$	4e 0e 12e 16e	 **2.11**	$2 \times Cl$ Ti $2 \times C_5H_5\bullet$	2e 4e 10e 16e
Co^{3+} $2 \times C_5H_5^-$	6e 12e 18e	 **2.12**	Co $2 \times C_5H_5\bullet$ Positive charge[a]	9e 10e −1e 18e

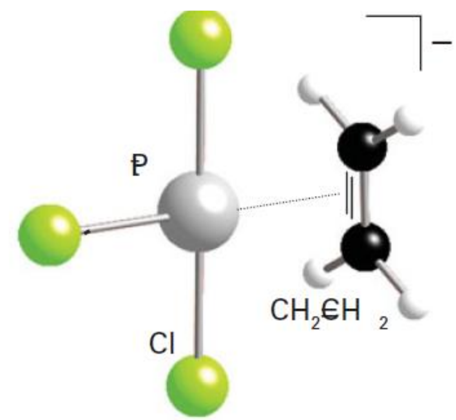
[a]To account for the positive ionic charge on the complex as a whole; for anions, the net charge is added to the total.



2 $Ni(CO)_4$



6 *trans*-$[IrCl(CO)(PPh_3)_2]$



1 $[PtCl_3(C_2H_4)]^-$